Two Modified Low-light Image Enhancement Methods

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Abstract

Low-light image enhancement (LLIE) aims to improve the visibility and restoring details in low-light images. In this study, two modified methods based on low-light enhancer from paired low-light instance (PairLIE) and LLIE with multi-stage residue quantization and brightness-aware attention (RQ-LLIE), respectively, are proposed. For the first modified method, a new illumination-guided attention block is added, and for the second modified method, an HSV correction net is added.

PairLIE – Introduction

PairLIE [1], an unsupervised LLIE network based on the Retinex theory, is trained by paired low-light images. For PairLIE, each image is decomposed into illumination and reflectance components. Assuming that the paired images share the same reflectance components, PairLIE will evaluate the lighting conditions from both images to reconstruct a normal-light image.

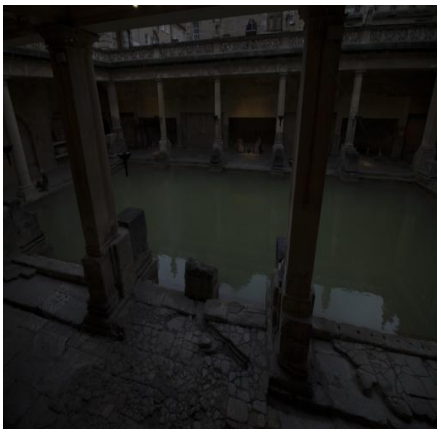
By the Retinex theory, an image I can be decomposed into illumination L and reflectance R:

$$I = L \circ R,$$

where $\circ$ denotes element-wise multiplication.



I_1



I_2

Fig. 1. A paired low-light images. I_1 and I_2 share the same reflectance

with differ illuminations.

The PairLIE architecture contains three modules P-Net, L-Net, and R-Net. P-Net is designed to remove inappropriate features of the input image. L-Net and R-Net are used to evaluate the illumination and reflectance components of the input image, respectively.

Aforementioned, a paired low-light images I_1 and I_2 share the same reflectance components with differ illuminations. These paired images are fed into PairLIE allowing the architecture to analyze the lighting conditions.

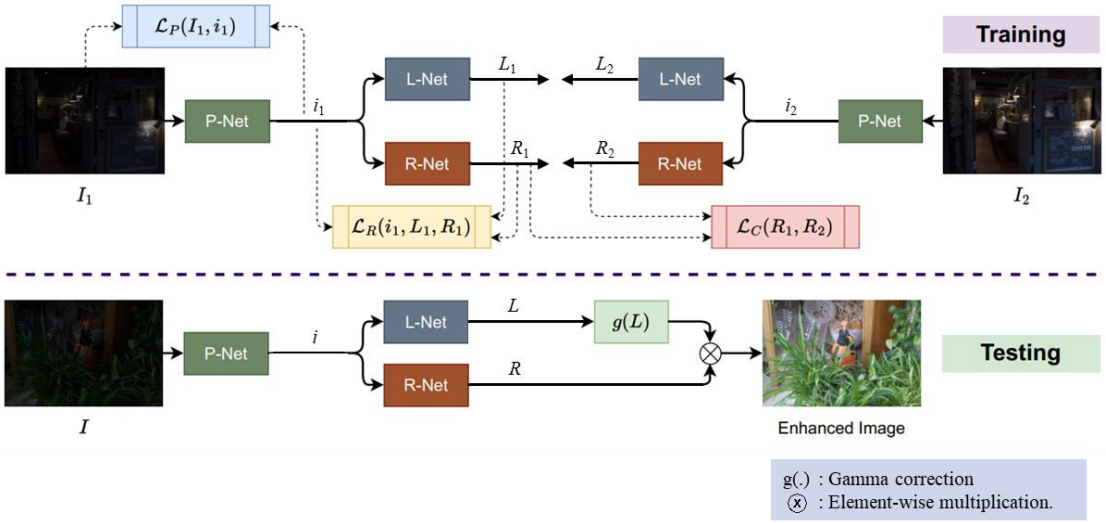


Fig 2. The PairLIE architecture in training phase (above) and testing phase (below).

PairLIE – Loss Function

To optimize the network, three loss functions are designed in PairLIE. The first is the projection loss $\mathcal{L}_p$ that measures the difference between I and i. The second is the reflectance consistency loss $\mathcal{L}_C$ calculated based on R_1 and R_2. The third is the Retinex loss $\mathcal{L}_R$ that restricts the decomposed components to satisfy the Retinex theory.

Projection Loss. Projection loss remove inappropriate features to help the decomposition more accurately. The following loss are designed to guide projection:

$$\mathcal{L}_p = \|I_1 - i_1\|_2^2$$

, where I is the original image and i is the expected to be noise free image.

Reflectance Consistency Loss. Reflectance consistency loss is based on Retinex theory the pair image shared the same reflectance components:

$$\mathcal{L}_C = \|R_1 - R_2\|_2^2$$

, where R_1 and R_2 refer to the reflectance components of paired low-light images. $\mathcal{L}_C$ enforces the network to predict to same reflectance components because low-light image pairs share the same objects.

Retinex Loss. Since Retinex theory decompose the image into illumination and reflectance component can be formulated as:

$$\begin{aligned}\mathcal{L}_R = &\|R \circ L - i\|_2^2 + \|R - i/stopgrad(L)\|_2^2 \\ &+ \|L - L_0\|_2^2 + \|\nabla L\|_1,\end{aligned}$$

where i denotes the projected image, L_0 refers to the initial estimation of illumination, ∇ represents the horizontal and vertical gradients.

Modified method 1

For the first modified method, we find that images have almost same light with the reference image but loss detail, so we pay more attention on the reflectance component

Two modules and one net are changed in the modified method , illumination estimator and illumination-guided attention block (IGAB), are used to assist R-Net to construct the reflectance component, changing the role of origin P-Net from produce noise free image directly to produce the noise subtracted by input image.

Illumination Estimator. Illumination estimator will analyze the light distribution in the image, and enhance the image illumination information,

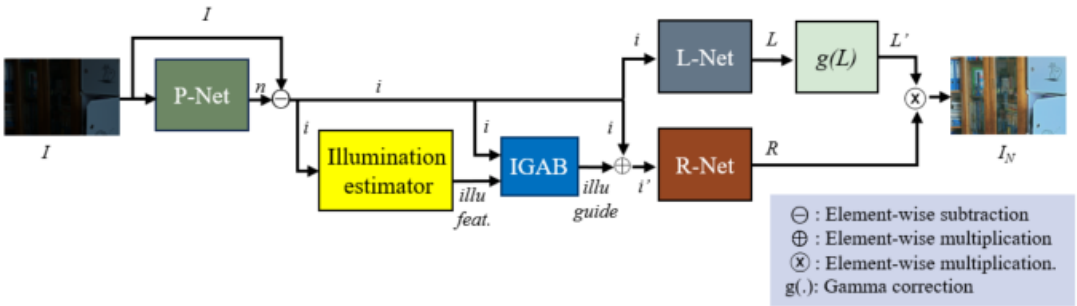


Fig. 3. Architecture of the first modified method.

IGAB. Illumination-guided attention block (IGAB) is proposed in Retinexformer [2], which is a basic block of illumination-guided transformer (IGT), designed as a corruption restorer.

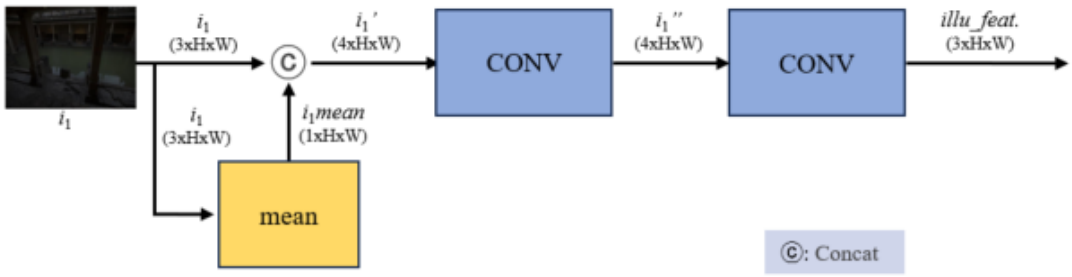


Fig 4. Illumination estimator architecture.

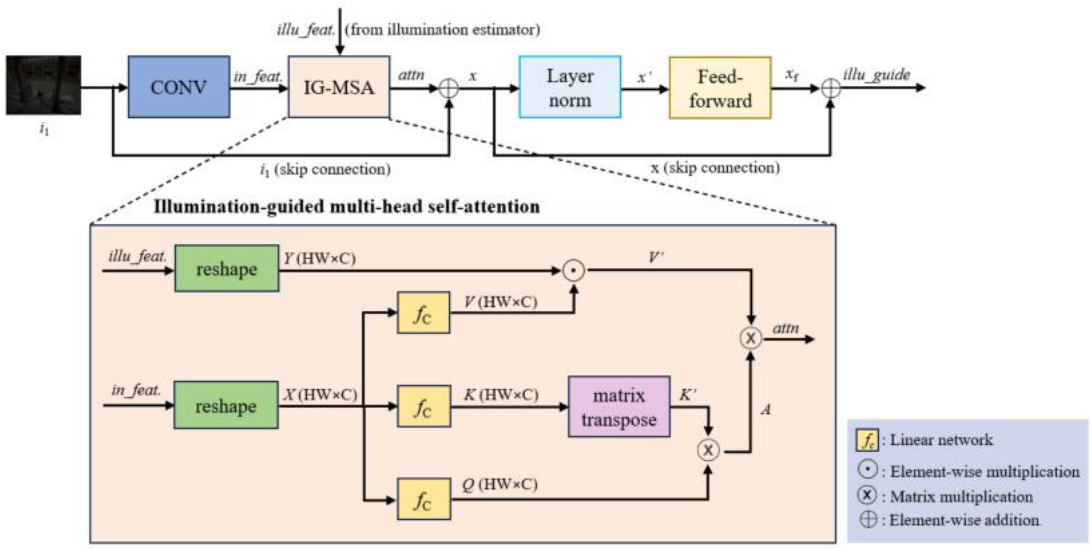


Fig 5. Illumination-guided attention block.

Modified Method 1 – Experimental Results

Compare to original method the higher PSNR/SSIM score show the image is closer to the reference.

Table 1. Performance comparisons between PairLIE and the first modified method.

Dataset	**LOL**		**SICE**	
Avg	PairLIE	Method 1	PairLIE	Method 1
PSNR (dB) (↑)	19.51	**19.66**	21.32	**21.54**
SSIM (↑)	0.736	**0.752**	0.840	**0.848**
LPIPS (↓)	0.248	**0.232**	0.216	**0.206**
Delta E (↓)	**10.80**	11.66	**7.835**	7.871



Input



Ground Truth



PairLIE



Modified Method 1



Fig 6. Visual comparisons between PairLIE and the first modified method.

RQ-LLIE – Introduction

RQ-LLIE [3], proposed by Liu, et al., is a novel low-light image enhancement method based on VQ-VAE [4] with a three-stage framework, featuring an expressive codebook and a query module.

RQ-LLIE is a supervised method with a three-stage framework. In stage 1, a codebook is constructed by residual quantization aiming to learn normal light features from the input

of normal-light images. In stage 2, a query module is designed to bridge the gap between low-light features and the normal-light codebook. In stage 3, a fusion branch is used to fuse low-light features and normal-light priors from the codebook and the query module.

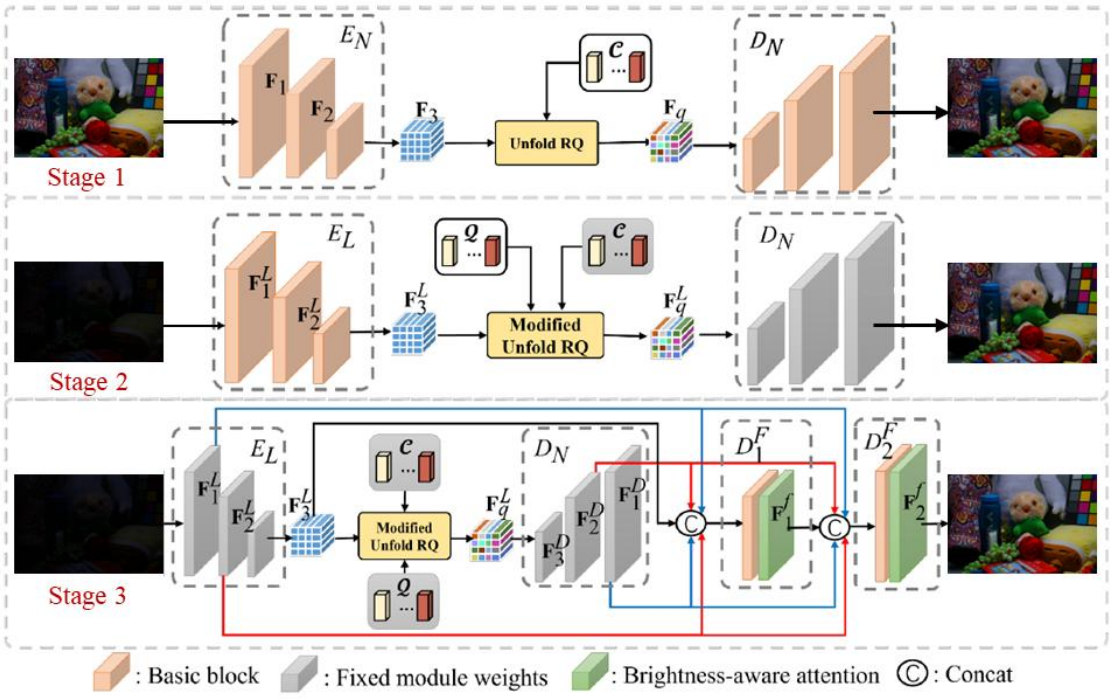


Fig 7. RQ-LLIE three-stage framework for low-light image enhancement.

Modified Method 2

From the visual comparisons between the results of RQ-LLIE and the ground truth, for some of the images, there is a gap between the references and the results in terms of saturation.

For the second modified method, to enhance the quality of output images, RQ-LLIE will be followed by an HSV correction net. Each input image will be converted to HSV color space. Then, the saturation and value components will be adjusted by S-Net and V-Net, respectively. Additionally, N-Net is used to convert the image back to RGB color space.

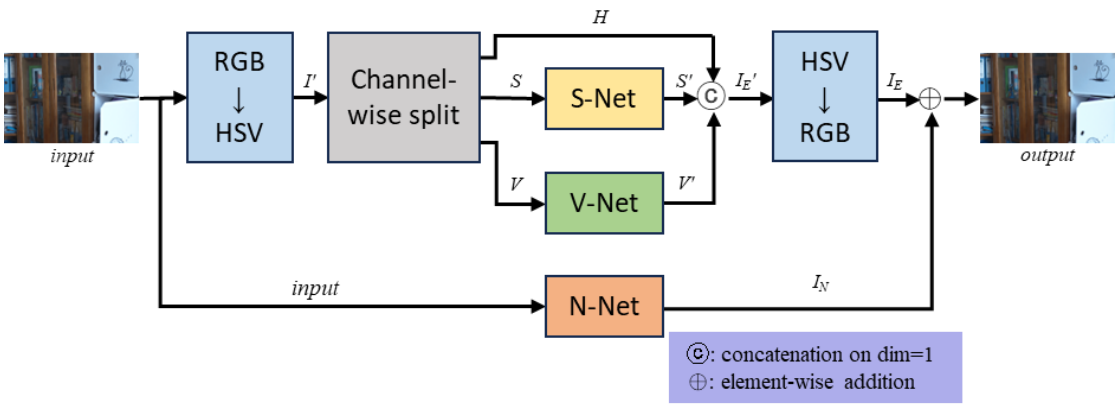


Fig 8. HSV correction net architecture.

Modified Method 2 – Loss Function

The HSV correction net is a supervised method, meaning the two images, the reference and the low-quality image, both of which be sent into the architecture. A loss function is designed to optimize the network:

$$\mathcal{L} = ||I_R' - I_{R_E}'||_2^2 + ||I_R - (I_{L_E} + I_{R_N})||_2^2 + ||I_{R_N} - I_{L_N}||_2^2$$

, where I_R and I_L represent the reference and the low-quality image, respectively.

Modified Method 2 – Experimental Results

Table 2. Performance comparisons between RQ-LLIE and the second modified method.

Dataset	LOLv1	
Avg	RQ-LLIE	Method 2
PSNR (dB) (↑)	25.2407	**25.5454**
SSIM (↑)	0.8551	**0.8611**
LPIPS (↓)	0.1211	**0.1158**
Delta E (↓)	7.0609	**6.9087**

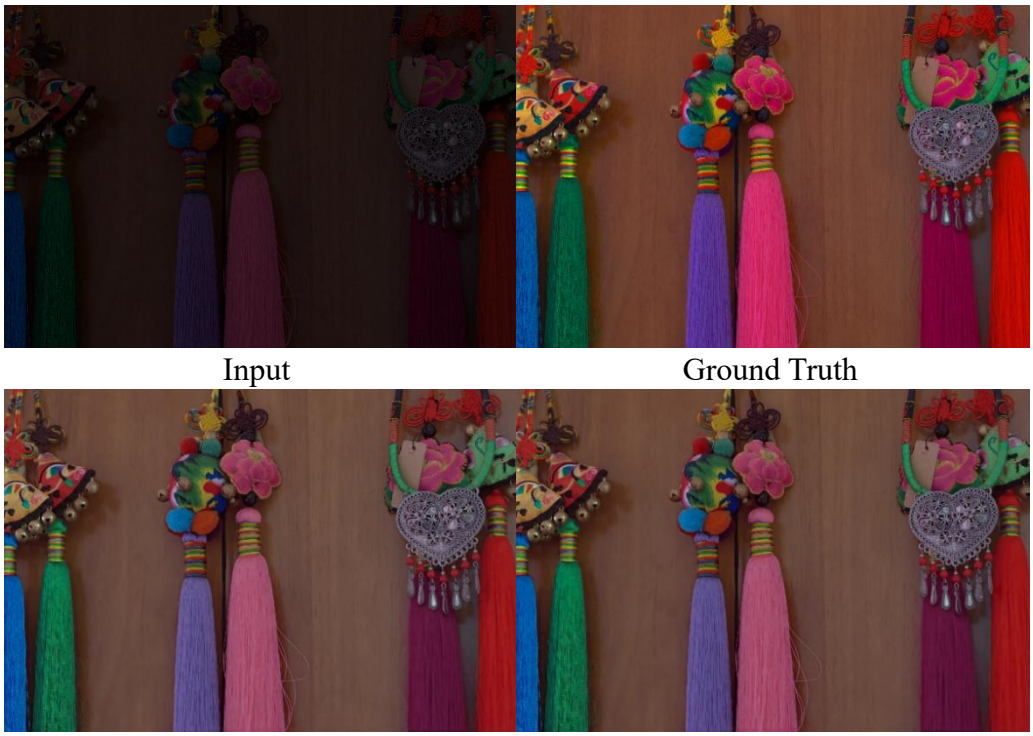


Fig 9. Visual comparisons between RQ-LLIE and the second modified method.

Concluding Remarks

In this study, two modified low-light image enhancement methods are proposed. Based on the experimental results obtained in this study, the performances of the two modified methods are better than those of two comparison methods, PairLIE and RQ-LLIE, respectively.

References

[1] Z. Fu, Y. Yang, X. Tu, Y. Huang, X. Ding, and K. Ma, “Learning a simple low-light image enhancer from paired low-light instances,” in *2023 IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, Vancouver, BC, Canada, 2023, pp. 22252-22261.

[2] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, “Retinexformer: one-stage Retinex-based transformer for low-light image enhancement,” in *2023 IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, Paris, France, 2023, pp. 12470-12479.

[3] Y. Liu, T. Huang, W. Dong, F. Wu, X. Li, and G. Shi, “Low-light image enhancement with multi-stage residue quantization and brightness-aware attention,” in *2023 IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, Paris, France, 2023, pp. 12106-12115.

[4] A. Van Den Oord, et al, “Neural discrete representation learning,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.